

Model Evaluation, Comparison & Improvement

Aspect-Based Sentiment Analysis on Laptop Reviews (Flipkart)

Overview

This report extends the earlier baseline sentiment classification model by comparing multiple algorithms, applying cross-validation, checking for overfitting, performing error analysis, testing a basic improvement, and selecting a final candidate model with justification. All models were trained and evaluated on the same train/test split and TF-IDF feature space for a fair comparison.

Models compared: Logistic Regression (carried forward from the earlier baseline), Multinomial Naive Bayes, and Linear SVM. Random Forest was excluded, tree-based models generally underperform on sparse, high-dimensional TF-IDF text features compared to linear approaches.

Single-Split Model Comparison

Model	Accuracy	Negative Precision	Negative Recall	Negative F1
Baseline (majority class)	0.82	0.00	0.00	0.00
Logistic Regression	0.87	0.60	0.80	0.69
Naive Bayes	0.87	0.91	0.34	0.50
Linear SVM	0.88	0.64	0.77	0.70

Naive Bayes reaches similar overall accuracy but has far weaker Negative recall (0.34), missing roughly two-thirds of actual negative reviews. This reflects that Naive Bayes has no built-in class imbalance handling, unlike Logistic Regression and Linear SVM, both trained with `class_weight="balanced"`. Linear SVM edges out Logistic Regression on every metric shown in this single split.

Cross-Validation (5-Fold, F1-macro)

A single train/test split can be misleading, so each model was additionally evaluated using 5-fold cross-validation on the training data, using F1-macro (average of Positive and Negative F1, weighted equally) as the scoring metric to avoid the majority class dominating the score.

Model	Mean F1-macro	Std Deviation
Logistic Regression	0.800	0.008
Naive Bayes	0.695	0.015
Linear SVM	0.807	0.006

Cross-validation confirms the single-split results: Linear SVM has both the highest mean F1-macro and the lowest variability across folds, indicating strong and consistent performance. Naive Bayes underperforms both other models and is not considered further as a final candidate.

Overfitting Check

Model	Train F1-macro	Test F1-macro	Gap
Logistic Regression	0.844	0.800	0.044
Linear SVM	0.890	0.813	0.077

Both models show reasonable train-test gaps with no strong evidence of overfitting. Logistic Regression generalizes slightly more tightly (gap of 0.044), while Linear SVM has a somewhat larger gap (0.077) but still achieves higher test performance overall, a mild trade-off rather than a concerning level of overfitting.

Error Analysis

A sample of false negatives (actual Negative reviews predicted as Positive by Linear SVM) was examined directly. Of 3,399 test reviews, 412 (12.1%) were misclassified, including 142 false negatives. Four recurring patterns emerged:

- 1. Mixed sentiment:** reviews open with positive words ("good", "awesome") before pivoting to the actual complaint. TF-IDF treats words largely independently of order, so word-level positivity can outweigh the structural cue that the complaint is the real message.
- 2. Truncated reviews:** several reviews are cut off mid-sentence, consistent with the "READ MORE" scraper artifact identified during earlier exploratory analysis (affecting ~3.7% of reviews). The negative content appears to have been cut off in these cases.
- 3. Seller/delivery complaints rather than product complaints:** phrases like "never buy from Flipkart" reflect a negative platform experience, not the laptop itself, yet carry the same binary label as product complaints. This motivates the planned move to aspect-level sentiment extraction.
- 4. Rating-text mismatch:** at least one case showed clearly positive text paired with a low star rating, likely label noise rather than a model error.

Feature Importance

As a linear model, Linear SVM's coefficients show which words most strongly influence predictions in each direction. Clear sentiment words dominate both lists (Positive: "awesome", "wonderful", "smooth"; Negative: "waste", "dead", "stuck"), confirming the model is largely learning genuine sentiment language.

However, some dataset-specific correlations were also observed: "msi" (a brand name) appears as a Positive indicator, likely because MSI laptops happen to have skewed positive reviews in this dataset rather than the word carrying inherent sentiment. Similarly, a specific processor model number appears as a Negative indicator. This suggests the model partly reflects patterns specific to this dataset and may not transfer perfectly to a different set of reviews.

Basic Tuning: Bigrams

Motivated by the mixed-sentiment pattern found in error analysis, bigrams (two-word phrase features, e.g. "not good") were added to the TF-IDF vectorizer and re-evaluated on Linear SVM using the same 5-fold cross-validation setup.

Configuration	Mean F1-macro	Std Deviation
Linear SVM (unigrams only)	0.807	0.006
Linear SVM (unigrams + bigrams)	0.815	0.006

Adding bigrams produced a small but consistent improvement (+0.007 mean F1-macro, same low variability), aligning with the error analysis finding that the unigram-only model struggled with short,

structurally important phrases. Bigrams are adopted for the final model.

Final Model Selection

Selected model: Linear SVM with bigrams (unigram + bigram TF-IDF features, `class_weight="balanced"`)

Predictive performance: Highest cross-validated F1-macro (0.815) among all models tested, with the best single-split test performance.

Stability: Lowest standard deviation across cross-validation folds (0.006), indicating consistent performance regardless of data split.

Business impact: Negative recall (catching actual negative reviews) is prioritized for this project, since missing a negative review is more costly than occasionally over-flagging a mixed review. Linear SVM's Negative recall (0.77) is close to Logistic Regression's (0.80) while offering better overall balance across metrics.

Interpretability: As a linear model, coefficients can be directly inspected, confirming the model largely learns genuine sentiment language, with some dataset-specific limitations noted.

Known limitations: struggles with mixed-sentiment reviews, truncated reviews from the scraper artifact, and reviews complaining about seller/delivery rather than product quality. Some learned patterns reflect dataset-specific correlations rather than pure sentiment. The 2-class (Positive/Negative) framing remains a simplification flagged for mentor discussion.

Conclusion

Linear SVM with bigrams is selected as the final candidate model for this phase, offering the strongest and most stable performance among the models tested while remaining interpretable. The patterns identified during error analysis, particularly mixed sentiment and seller-versus-product complaints, directly inform the next phase of this project: moving from overall sentiment classification toward aspect-level sentiment extraction.